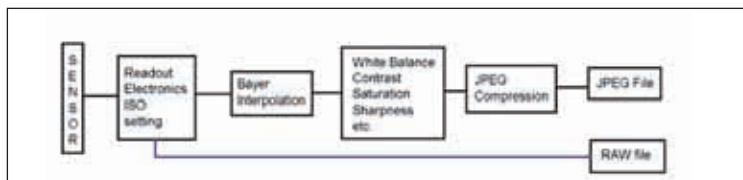


Image formats – RAW or JPEG?

Ah, this is a question every photographer has faced at some point or the other. Countless discussion threads and references cover this debate, concerning issues ranging from basics like size and quality to advanced technicalities like colour bits per channel, compression etc. In simple terms, RAW data is the output from every original red, green or blue sensitive pixel of the image sensor, after being read by array electronics and having passed through analog to digital converters. In comparison to storing data as a JPEG file, RAW file data goes through the Bayer interpolation filter and is modified as per set parameters like white balance, sharpness etc before being subjected to JPEG compression and then stored, ready for viewing and printing immediately after the shot. File sizes are smaller in JPEG and can be interpreted by many programs giving it a basic advantage over RAW but a counter-effect is the loss in quality proportionate to the compression. The exact way in which JPEG images are processed within the camera varies from model to model, depending on the camera setting for parameters like colour temperature and exposure. It also involves internal processing for characteristics such as contrast, brightness, noise reduction and sharpness. You may come across terms like “dynamic range” commonly thrown around when discussing the RAW vs JPEG topic. This is simply an indicator of the amount of tonal range detail from the darkest shadows to the brightest highlights. Because of the image compression when storing JPEG, a significant amount of initial information and detail is discarded which can't be recovered later, leading to reduced dynamic range in JPEGs as compared to RAW.

A RAW file is not exactly an image per se, and to view it typically in proprietary formats requires a special (and commonly available) software. Most notably, it's an uncompressed file. What this means is that a 10-megapixel camera produces a 10MB RAW file which has the complete data from the camera's sensors without any loss and with a high dynamic range.



Raw vs jpeg